Rank-27 Congruence Scope Repair after the Vivify Audit

Tactix engineering handoff

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Objective and outcome

The immediate objective was to continue the immutable Tier-D ranks 21–40 campaign in UNSAT-first order at rank 27. The user correctly rejected the formula-range Vivify disable as an instance-shaped policy; that repair was completed in the preceding increment and is documented in `report-to-andrew/2026-08-20-2357-vivify-admission-rank24/`. This increment then compared Tactix and Kissat from rank 27's latest common semantic state, including Vivify scheduling, candidate order, and rule interleaving, while looking for the earliest portable divergence.

The earliest divergence turned out to precede Vivify and search. Tactix silently excluded AND gates above arity 64 from opening Congruence; both Kissat and CaDiCaL ship the same option at 1,000,000. Raising only that scope value restores exactly 132 opening equivalence classes that Tactix was missing. A new arity-65 unit fixture pins the first old exclusion.

The repair passes the full required correctness and iteration guard, but rank 27 itself remains open: the repaired candidate times out UNKNOWN at 1,200.004 seconds. Thus this is a verified semantic-parity repair, not a Tier-D closure claim.

Frozen identities and non-negotiables

Rank 27 is the immutable UNSAT instance `tierD24-26/5911653e4edadec391654c8f06353f9a-oski15a01b06s_opt.cnf.xz`, compressed SHA-256 `5435b1b2...bb4e9`. Frozen Kissat solves it in 278.187 seconds. That reference performance result was not rerun; only a pinned-source diagnostic build was instrumented.

The accepted predecessor binary is `b6c991e0...fb6f`, inherited from the nine checked ranks-21–40 closures. The repaired candidate is `46a483ee...699`, built from commit `7b527fbc...204b` plus source diff `b2885a47...0c81`. The solver contains no benchmark name, hash, rank, filename, deadline, selected literal, per-instance arm, or locally tuned threshold. The value 1,000,000 is the shared shipped reference default already recorded in the repository's scheduling-parity ledger.

Localization from consumer to producer

The retained pre-repair traces agreed through decision 69,892. At decision 69,893, Tactix selected variable 392,211 and Kissat selected 392,214 at the same activity score. This made stable-heap shape the first visible search boundary. The consumer was then ruled out:

- conflict 1532, its bumped-variable stream, and its top-32 heap were exact;
- all 35,887 heap reinsertions through conflict 1532 matched in order;

- Tactix and Kissat use the same strict activity comparison, equal-score stopping rule, and left-child preference.

The opening producer state was already different. Kissat's Congruence merged 58,722 variables: 55,658 through AND gates, 2,995 through XOR gates, and 69 through paired binary equivalences. Tactix merged 58,590: 55,526 AND, 2,995 XOR, and 69 binary. The entire 132-variable population difference was therefore in AND processing, and it explained the later heap-shape tie without requiring a downstream tie breaker.

Diagnostic Slurm job `1636967` measured Tactix's gate-kind funnel. Job `1636968` instrumented the unchanged Kissat semantics and showed that ITE-to-AND conversions contributed zero of Kissat's opening AND merges. Job `1636970` dumped both final representative partitions; they differed on exactly 132 variables. The first witness was `93231 == -93230` in Kissat while Tactix left 93231 singleton.

The defining clauses form matching 72-input gates. Selected-gate job `1636972` showed that Tactix never created the high-fan-in gate: its `CONG_ARITY_LIMIT = 64` test rejected the base clause before the worklist. This matched correction C7 in `docs/scheduling-parity-ledger.md`, which had already identified the 64-versus-1,000,000 value gap and named the exact reference-parity repair.

Implementation decision

The production change is one scope value in `src/rules/congruence/congruence.cpp`: AND arity 64 becomes 1,000,000. Tactix retains its separately documented mid-round work budget as the effort governor. The gate representation, extraction order, union-find, proof construction, scheduling, and public rule surface do not change.

`tests/test_congruence.cpp` now builds two congruent 65-input AND gates and requires their outputs to merge. Arity 65 is deliberately the first old exclusion. The fixture is independent of rank 27 and would catch a future silent narrowing of the reference-wide opportunity.

Correctness and matched iteration guard

Release passes 72/72 tests, including native LRAT and DRAT micro-proofs and the new wide-AND regression. No unit test or fuzz check exposed a soundness defect or crash, so the development plan's mandatory Claude repair-delegation trigger did not fire.

Matched job `1636973` ran the accepted predecessor and repaired candidate in the same window on all 28 Tier-B fixtures and the predeclared fixed five Congruence-relevant Tier-C fixtures. The exact contract was two CPUs, 16 GiB process memory, 1,200-second solver timeout, no node pin, and no array throttle.

Arm	Tier B	Tier C	checked total	PAR-2 / wall
Accepted predecessor	28/28	5/5	33/33	2228.698 s
Repaired candidate	28/28	5/5	33/33	2239.817 s

The candidate ratio is 1.004989. There is no solve loss: each arm returns 17 SAT and 16 UNSAT answers. All 34 SAT models and 32 UNSAT proofs across both arms verify. Candidate peak RSS is 3.228 GiB. Every task reports deleted scratch and derived artifacts, and the aggregate finds no certificate residue. The campaign is `benchmarks/runs/20260821-0035-BC-congruence-arity/`.

Post-repair diagnostic job `1637040` directly confirms the corrected opening counts: 55,658 AND merges, 2,995 XOR merges, and 69 paired-binary merges, 58,722 total. Its deliberate 15-second timeout deletes the partial proof and leaves no residue. The compact campaign is `benchmarks/runs/20260821-0125-tierd-rank27-repaired-congruence-count/`.

Rank-27 result

Only after the matched guard passed was the immutable witness submitted as job `1637039`. The repaired candidate reaches the 1,200-second solver limit and returns UNKNOWN. The runner observed a 128,974,848-byte partial DRAT file, deleted it with the scratch directory, and published zero cleanup residue. Scheduler peak RSS was 496,616 KiB. The campaign is `benchmarks/runs/20260821-0052-tierd-rank27-congruence-arity/`.

This result means the semantic repair should land, but rank 27 must not be listed as closed. The cumulative accepted Tier-D set remains ranks 21, 23, 24, 25, 28, 29, 34, 38, and 39.

Limitations, risks, and next action

The repaired opening Congruence partition now matches Kissat in the dimension that caused the previous search split, but no claim is made that all later semantic state is exact. The old decision-69,893 boundary depended on the 132-variable population gap and is obsolete for the repaired source. Rank 27 also shows that restoring a missing proof opportunity need not improve the overall search trajectory by itself.

The next concrete action is to build a fresh semantic-trace pin and compare it with the retained diagnostic Kissat build beginning after the corrected 58,722-merge opening Congruence/Substitute state. The first new portable semantic, scheduling, or representation divergence should be localized before any further edit. Any new candidate again owes the full Tier-B plus predeclared Tier-C guard before another rank-27 witness.

The durable technical narrative is `benchmarks/reports/2026-08-21-tierd-rank27-congruence-arity.md`. Current handoff state is in `PROGRESS.md`; the repaired reference scope and reopening evidence are in `docs/scheduling-parity-ledger.md` and `docs/implementation-notes.md`.